JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI MURUGAN AGENCIESIOC DEALERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

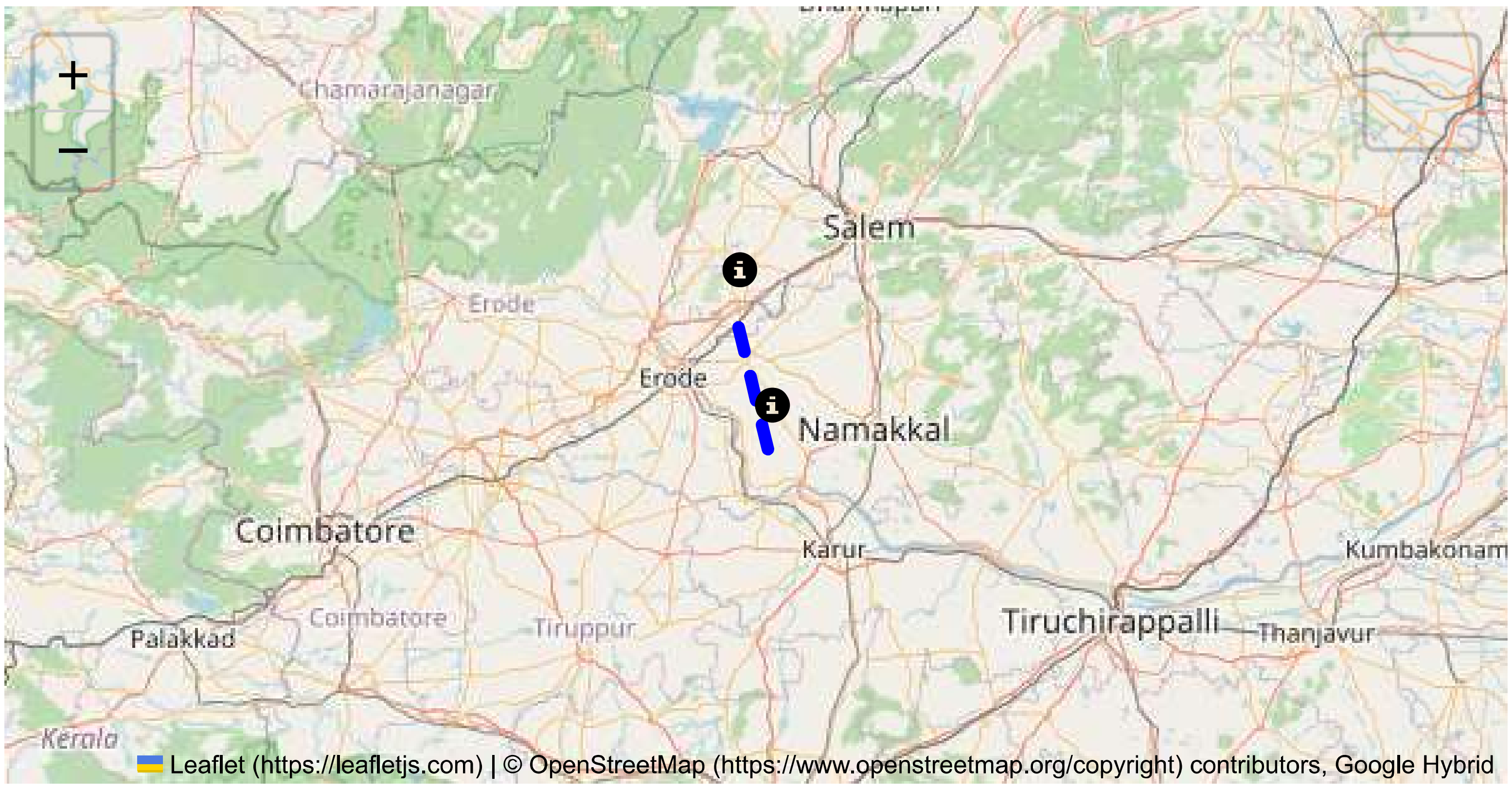
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

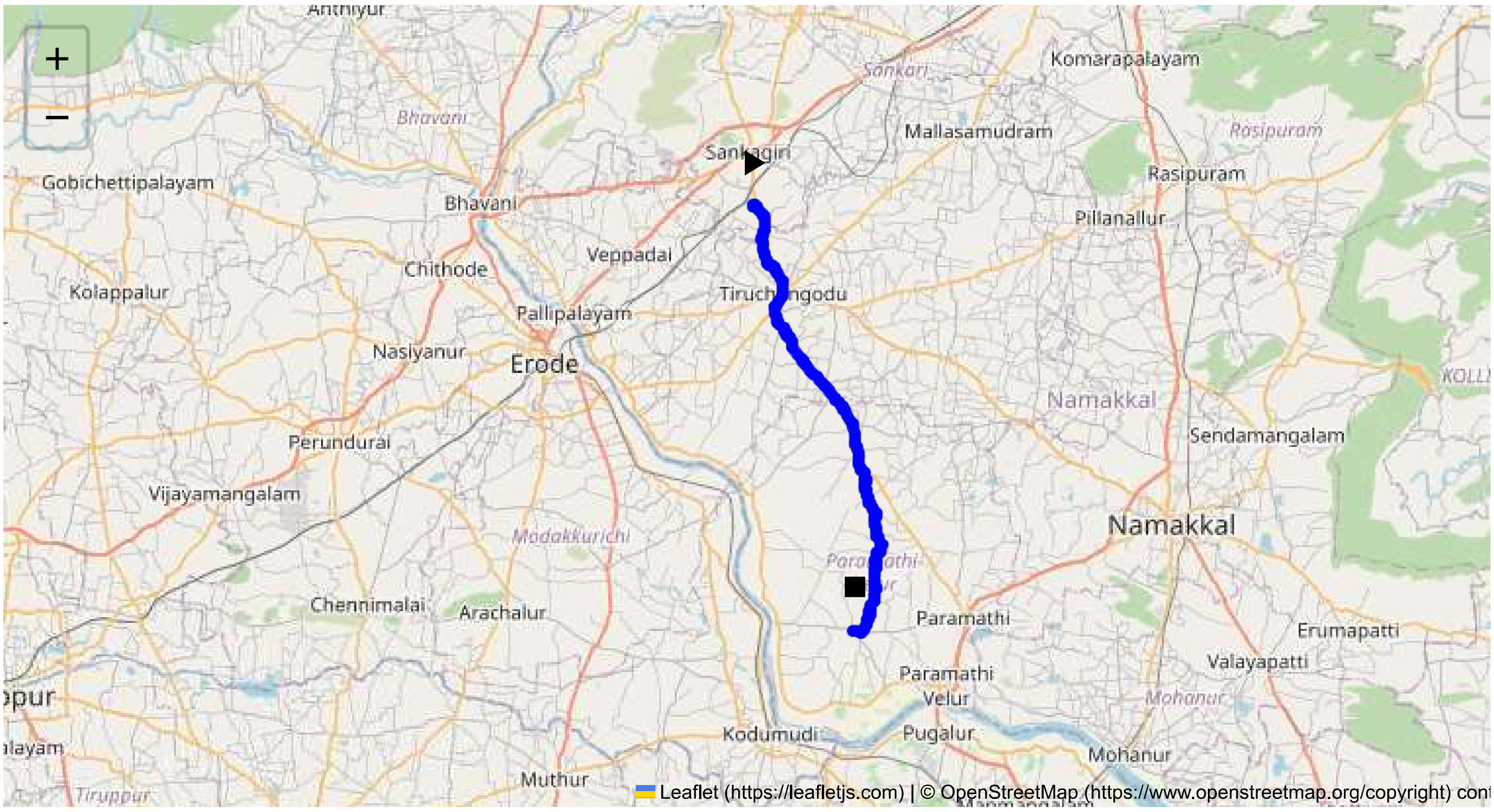
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 38.36 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.146426, 77.944138)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map The route from CVQF+23W, Sangagiri, to 4WWV+JHX, Kandhan Nagar, spans approximately 38.36 kilometers and follows a primarily rural path. The journey typically takes just under one hour for heavy vehicles transporting hazardous materials. The route likely includes a mix of state highways and local roads, characterized by smaller towns and agricultural landscapes.

2. Weather Conditions and Potential Weather-Related Hazards The typical weather conditions in Tamil Nadu include a hot and humid climate with a pronounced monsoon season. The southwest monsoon (June to September) and northeast monsoon (October to December) can bring heavy rainfall, which may lead to waterlogged roads, reduced visibility, and the potential for landslides in hilly areas. Cyclonic activities may also pose significant threats during these periods.

3. Traffic Patterns and Congestion Traffic patterns in this region show relatively low volume outside urban centers. Peak hours coincide with local business hours, usually between 8-10 AM and 5-7 PM. Key congestion-prone areas may include town centers or where major state highways intersect. However, rural roads may experience slower-moving agricultural vehicles at all times.

4. Road Quality and Infrastructure The road quality varies with segments of well-maintained highways interspersed with narrower local roads that might be less maintained. Look for potential issues such as potholes, limited signage, and inadequate lighting which can pose significant challenges for heavy vehicles.

5. Suggestions for Alternative Routes In case of an emergency or major disruption, consider alternate larger highways or bypasses that parallel the primary route, even if these might add some distance. Utilizing Highway 44 or other state highways could offer more reliable infrastructure and emergency services.

6. Local Regulations Affecting Hazardous Material Transport Transporting hazardous materials is subject to regulations specific to Tamil Nadu which include state permits, specific transit times (often restricted during night hours), and designated rest areas. Ensure compliance with state regulations, including permits and proper vehicle documentation.

7. Historical Incidents There have been periodic incidents involving heavy vehicles in Tamil Nadu, often attributed to driver fatigue, overloading, or inadequate vehicle maintenance. Few might specifically involve hazardous materials, highlighting the importance of strict adherence to safety protocols.

8. Environmental Considerations and Sensitive Areas The route may traverse agricultural zones vital to local economies and ecosystems. It is crucial to prevent spillage of hazardous materials that could impact soil and water quality. Monitoring wildlife crossings and protected areas is also crucial.

9. Communication Coverage While urban areas will have robust communication networks, rural stretches may experience intermittent coverage, especially between towns. Mapping these zones in advance is advisable to ensure drivers are prepared and alternative communication measures are in place.

10. Estimated Emergency Response Times Response times can vary significantly based on proximity to urban areas, possibly ranging from 15 to 40 minutes in rural stretches. Ensuring quick communication with emergency services and knowing the nearest facilities in advance can mitigate delays.

12. Overall Summary of Risk Assessment The route presents moderate risk primarily due to fluctuating road conditions, weather-related disruptions, and variable communication coverage. Thorough planning, detailed vehicle maintenance, and strict compliance with local regulations are essential to minimizing potential hazards. Alternative routes should be preplanned for emergencies, and continuous real-time communication systems should be used to ensure the safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	High	11.25570, 77.95157	15 KM/Hr
8	Turn	Medium	11.25015, 77.95213	30 KM/Hr
9	Turn	Medium	11.24999, 77.95225	30 KM/Hr
10	Turn	Medium	11.24966, 77.95293	30 KM/Hr
11	Turn	Medium	11.24542, 77.95155	30 KM/Hr
12	Turn	Medium	11.24528, 77.95152	30 KM/Hr
13	Turn	Medium	11.24495, 77.95186	30 KM/Hr
14	Turn	Medium	11.22160, 77.95876	30 KM/Hr
15	Turn	Medium	11.22122, 77.95914	30 KM/Hr
16	Turn	Medium	11.20578, 77.96269	30 KM/Hr
17	Turn	Medium	11.20500, 77.96392	30 KM/Hr
18	Turn	Medium	11.20483, 77.96397	30 KM/Hr
19	Turn	Medium	11.19990, 77.96177	30 KM/Hr
20	Turn	Medium	11.19979, 77.96118	30 KM/Hr
21	Turn	High	11.19389, 77.95912	15 KM/Hr
22	Turn	Medium	11.19323, 77.96032	30 KM/Hr
23	Turn	Medium	11.15787, 77.95474	30 KM/Hr
24	Turn	Medium	11.15661, 77.95555	30 KM/Hr
25	Turn	High	11.14502, 77.94957	15 KM/Hr
26	Turn	High	11.14616, 77.94394	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium